복사전달 특강
Special Topics in Radiative Transfer

Lecture 2
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Source Function for Multiple Processes

- When multiple processes contribute to local emission and extinction, the RT equation and the total source function are given by

$$\frac{dI_\nu}{ds} = -\left(\sum \alpha_\nu\right) I_\nu + \sum j_\nu \quad \Rightarrow \quad d\tau_\nu = \sum \alpha_\nu ds \;\; \text{and} \;\; S_\nu^{\text{tot}} = \frac{\sum j_\nu}{\sum \alpha_\nu}$$

- For example, the source function at a frequency ν within a spectral line is

$$S_\nu^{\text{tot}} = \frac{j_\nu^c + j_\nu^l}{\alpha_\nu^c + \alpha_\nu^l} = \frac{S_\nu^c + \eta_\nu S_\nu^l}{1 + \eta_\nu}$$

Here, $\eta_\nu \equiv \dfrac{\alpha_\nu^l}{\alpha_\nu^c}$ = line-to-continuum extinction ratio

$S_\nu^c \equiv \dfrac{j_\nu^c}{\alpha_\nu^c}$ = continuum source function

$S_\nu^l \equiv \dfrac{j_\nu^l}{\alpha_\nu^l}$ = line source function

[Note]

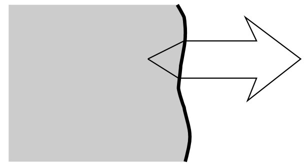
$I_\nu,\ F_\nu,\ j_\nu,\ S_\nu$ $\Rightarrow$ The subscript ν implies measurement per frequency interval.

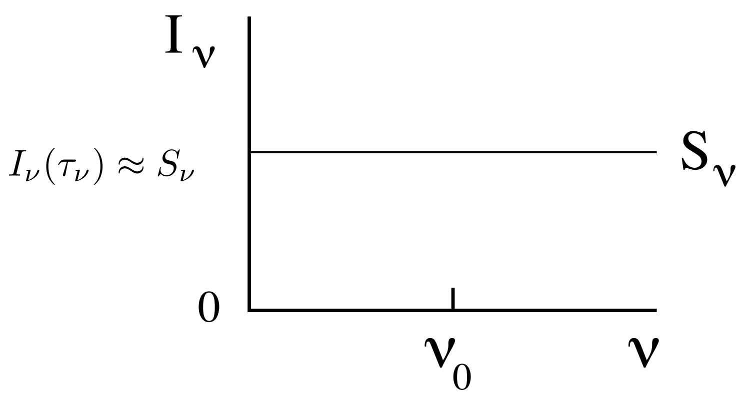
$\alpha_\nu,\ \sigma_\nu,\ \kappa_\nu$ $\Rightarrow$ The subscript ν simply expresses frequency dependence.

Spectral Lines from a Homogeneous Object with a constant Source Function

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + S_\nu\left(1 - e^{-\tau_\nu}\right)$$

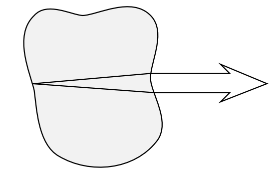
$\tau_\nu(D) >> 1$

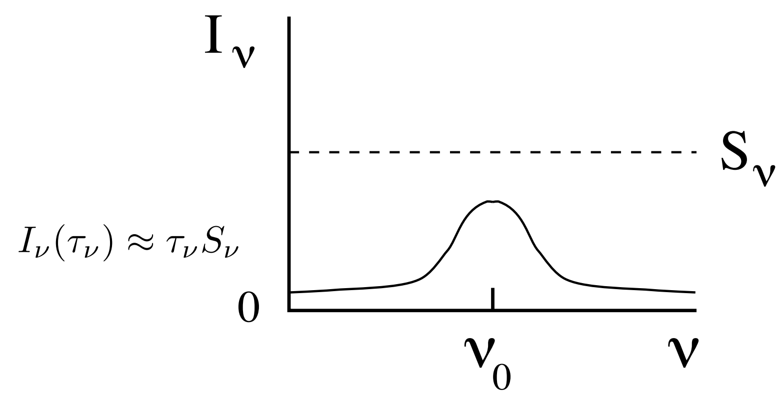




No lines emerge when the object is optically thick.

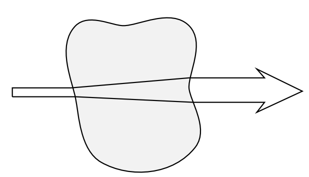
$\tau_\nu(D) < 1$
$I_\nu(0) = 0$

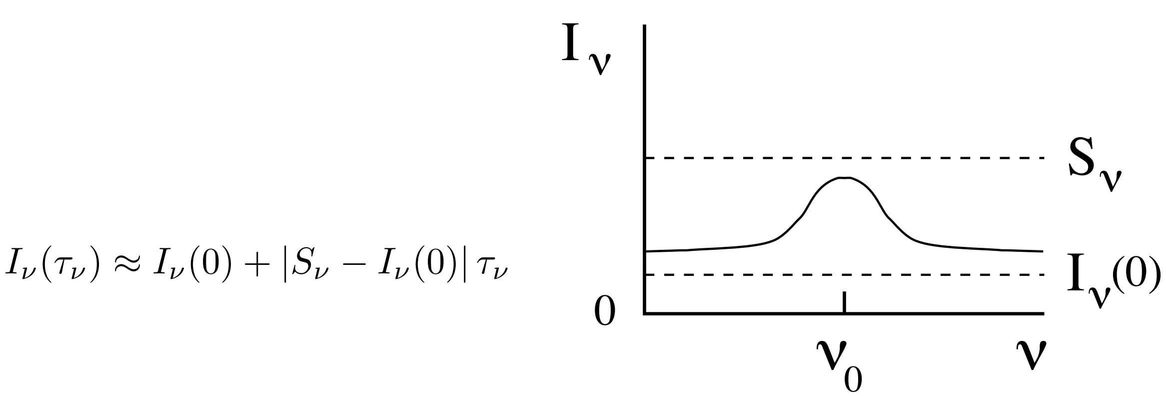




When the object is optically thin and not back-lit ($I_\nu(0) = 0$), emission lines emerge.

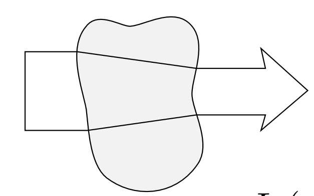
$\tau_\nu(D) < 1$
$I_\nu(0) < S_\nu$

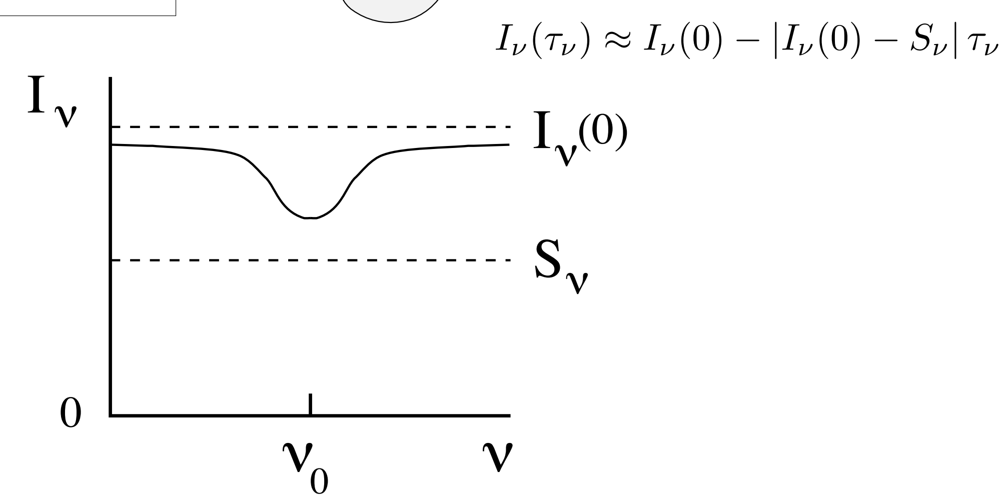




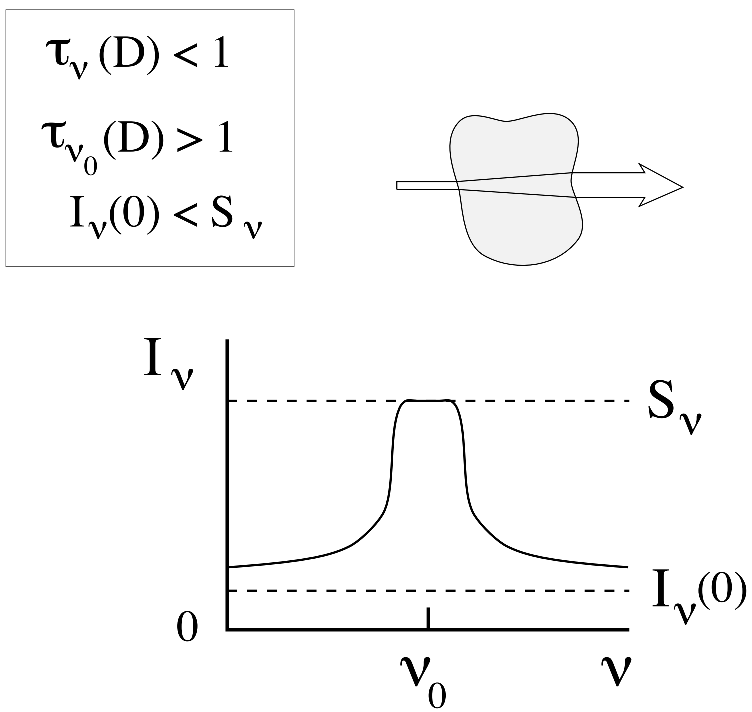
When the object is optically thin and illuminated with $I_\nu(0) < S_\nu$, emission lines emerge.

$\tau_\nu(D) < 1$
$I_\nu(0) > S_\nu$

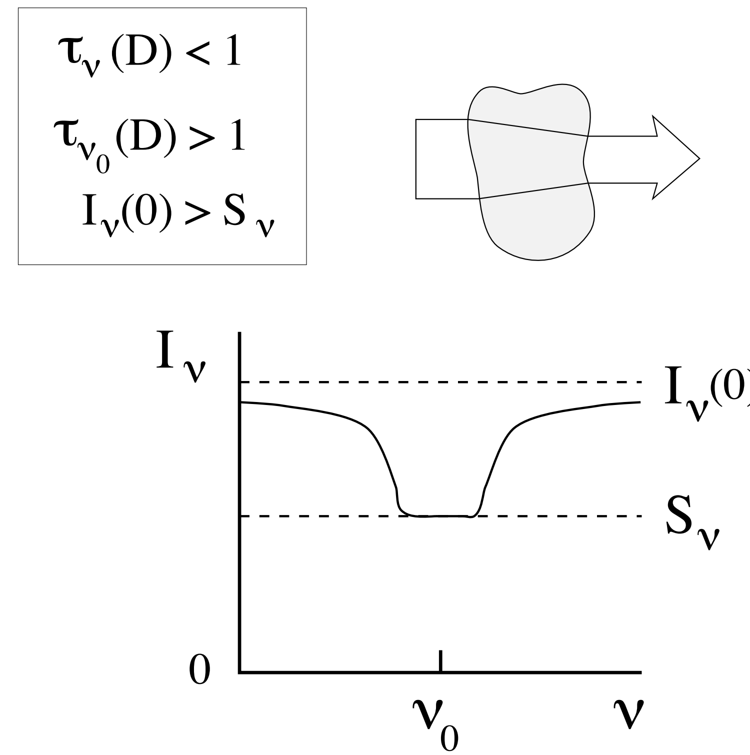




Absorption lines emerge only when the object is optically thin and $I_\nu(0) > S_\nu$.



The emergent lines saturate to $I_\nu(0) \approx S_\nu$ when the object is optically thick at line center (ν_0), but optically thin at $\nu \neq \nu_0$.

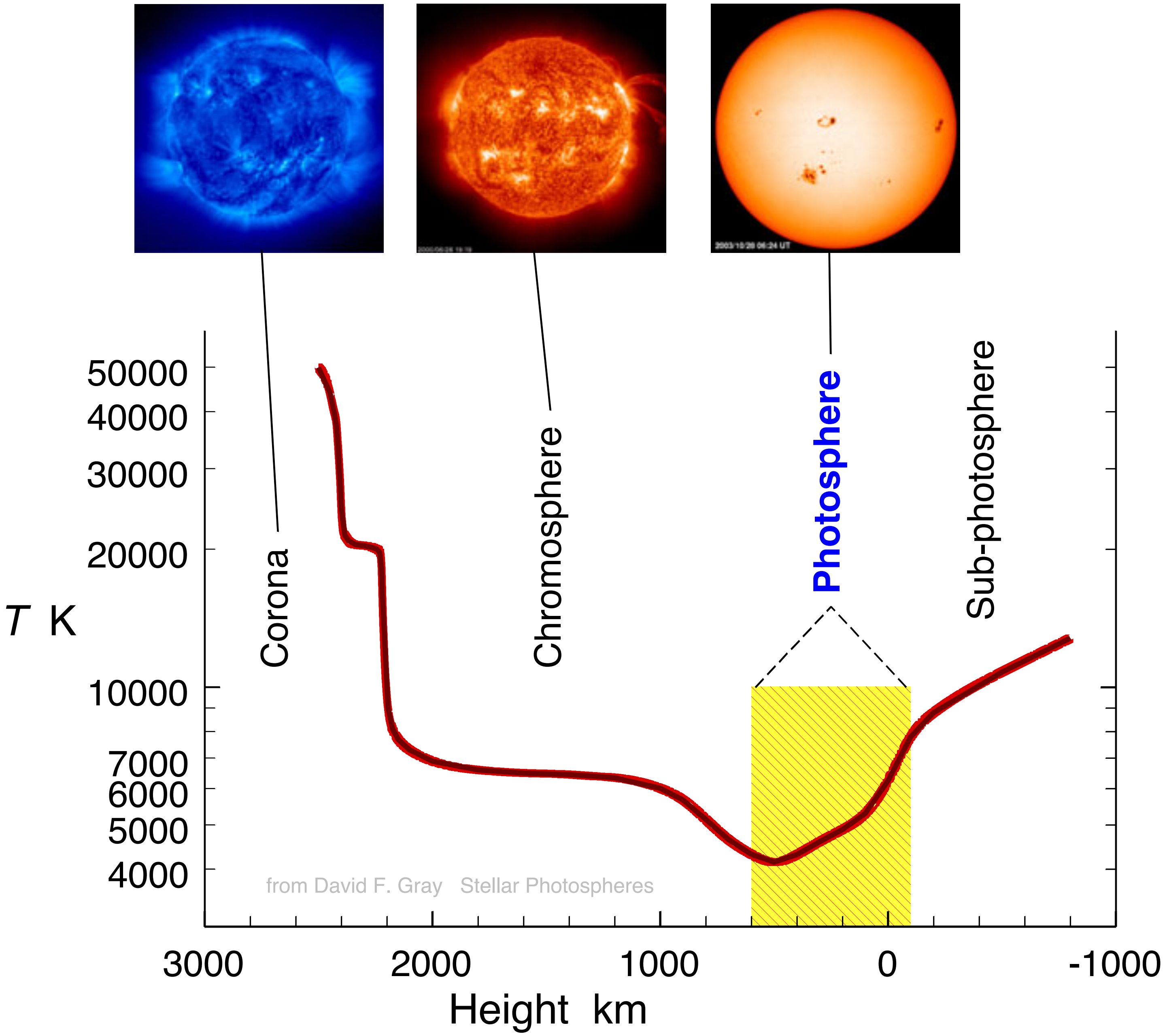


The emergent lines saturate to $I_\nu(0) \approx S_\nu$ when the object is optically thick at line center (ν_0), but optically thin at $\nu \neq \nu_0$.

$$\begin{aligned} I_\nu(\tau_\nu) &\approx S_\nu && \text{at } \nu \approx \nu_0 \\ &\approx I_\nu(0) + |S_\nu - I_\nu(0)|\,\tau_\nu && \text{at } \nu \neq \nu_0 \end{aligned}$$

$$\begin{aligned} I_\nu(\tau_\nu) &\approx S_\nu && \text{at } \nu \approx \nu_0 \\ &\approx I_\nu(0) - |I_\nu(0) - S_\nu|\,\tau_\nu && \text{at } \nu \neq \nu_0 \end{aligned}$$

Stellar Atmosphere - Photosphere



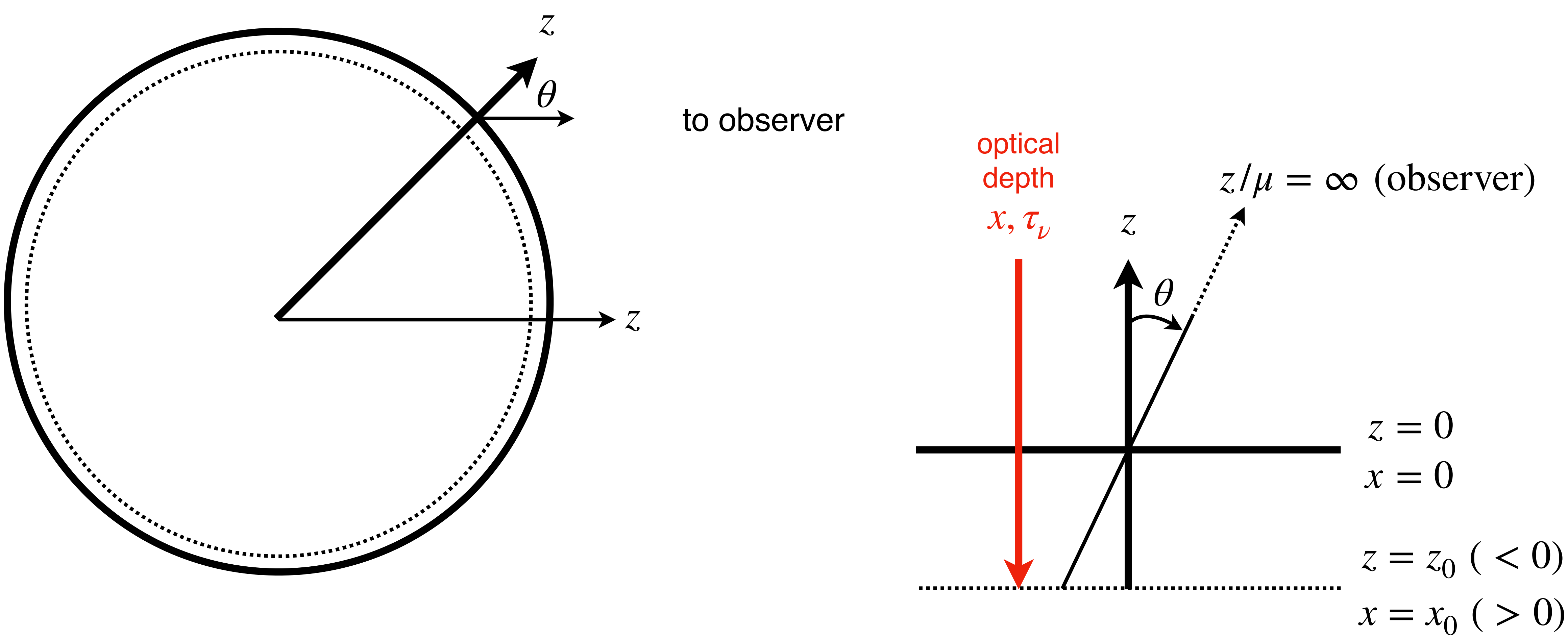
The temperature in the outer layers of the Sun, as a function of geometrical height in km.
[left] inner corona Fe IX/X 171Å line, [center] chromosphere at 304Å,
[right] photosphere in white light with large sunspots

Images courtesy of SOHO consortium (ESA & NASA); D. F. Gray

- The photosphere is the Sun's visible, lower surface with a temperature of ~5,500 K.
    - Light from the sub-photospheric layers does not penetrate to the surface, so we cannot see those layers.
- The chromosphere lies directly above it, appearing as a reddish, hotter layer (~4,400 — 25,000 K) that is only visible during eclipses or via specialized filters.
- The corona is the outermost layer extending 10^6 km into space, with temperatures ranging from 1 to 3×10^6 K.

Radiative Transfer in Stellar Atmospheres

- In the context of stellar atmospheres, one often adopts axial symmetry with the z-axis radially outward along the axis of symmetry (perpendicular to the surface of a spherical star consisting of horizontally homogeneous shells). *The optical depth is measured along the viewing direction (x-axis in the figure below).*
- The geometrical thickness of the photosphere in most stars is very small compared to the stellar radius.
  - The solar photosphere is ~700 km thick, or ~0.1% of the solar radius (~6.96×10^5 km).
  - Under these conditions, ***a plane-parallel approximation can be made.***



- The viewing angle μ is defined by $\mu = \cos\theta$ where θ is the angle between the line of sight and the z-axis.
- In stellar atmospheres, the total optical thickness, measured from the stellar center, along the line of sight is far too large to be any interest. Therefore, we define the optical depth τ_ν measured along the radial line of sight from the observer's location at $x = -\infty$ $(z = \infty)$. Then, the optical depth for a geometrical location with $x = x_0$ $(z = z_0)$ is given by

$$\tau_\nu(x_0) = \int_{-\infty}^{x_0} \alpha_\nu dx \quad \Rightarrow \quad \int_0^{x_0} \alpha_\nu dx \quad \text{as } \alpha_\nu = 0 \text{ for } x < 0$$

- For a frequency within a spectral line, the total optical depth is given by

$$d\tau_\nu^{\text{total}} = \left(\alpha_\nu^c + \alpha_\nu^l\right) dx = (1 + \eta_\nu)\, d\tau_\nu^c$$

- The radiative transfer equation in plane-parallel geometry is then given by

$$ds = -\frac{dx}{\mu} \quad \Rightarrow \quad \boxed{\mu \frac{dI_\nu}{d\tau_\nu} = I_\nu - S_\nu}$$

- Formal solution:

$$\frac{dI}{d(\tau/\mu)} = I - S$$

$$\frac{d}{d(\tau/\mu)}\left(e^{-\tau/\mu} I\right) = -e^{-\tau/\mu} S$$

$$I(\tau) = I(\tau^b) e^{-(\tau^b - \tau)/\mu} - \int_{\tau^b}^{\tau} S(t) e^{-(t-\tau)} dt/\mu$$

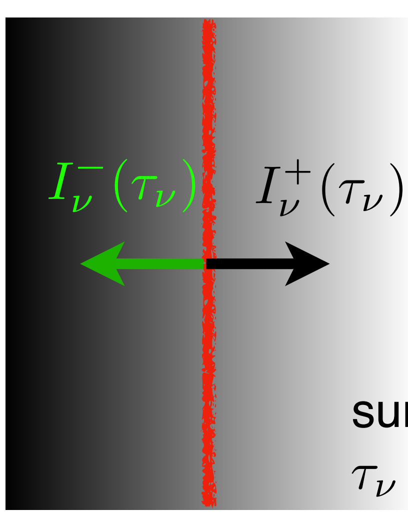
  - The inward directed intensity ($\mu < 0$) and outward directed intensity ($\mu > 0$) are respectively given by

$$\boxed{\begin{aligned} I_\nu^-(\tau_\nu, \mu) &= -\int_0^{\tau_\nu} S_\nu(t_\nu) e^{-(t_\nu - \tau_\nu)/\mu} dt_\nu/\mu \quad \text{for } \mu < 0 \\ I_\nu^+(\tau_\nu, \mu) &= +\int_{\tau_\nu}^{\infty} S_\nu(t_\nu) e^{-(t_\nu - \tau_\nu)/\mu} dt_\nu/\mu \quad \text{for } \mu > 0 \end{aligned}}$$

boundary conditions:

$$I^-(\tau^b, \mu) = 0 \quad \text{at } \tau^b = 0$$

$$e^{-\tau^b} = 0 \quad \text{at } \tau^b = \infty$$



interior
$\tau_\nu = \infty$

surface
$\tau_\nu = 0$

Eddington-Barbier Approximation

- Suppose that the source function can be approximated as a polynomial in optical depth:

$$S_\nu(\tau_\nu) = \sum_{n=0}^{\infty} a_n \tau_\nu^n = a_0 + a_1\tau_\nu + a_1\tau_\nu^2 + \cdots$$

- Then, the emergent intensity at the stellar surface ($\tau_\nu = 0,\ \mu > 0$) is given by

$$\begin{aligned} I_\nu^+(0,\mu) &= \int_0^\infty S_\nu(t_\nu) e^{-t_\nu/\mu} dt_\nu/\mu \\ &= \sum_{n=0}^{\infty} n! a_n \mu^n = a_0 + a_1\mu + 2a_2\mu^2 + \cdots \end{aligned}$$

- Truncation of both expansions after the first two terms produces the Eddington-Barbier approximation:

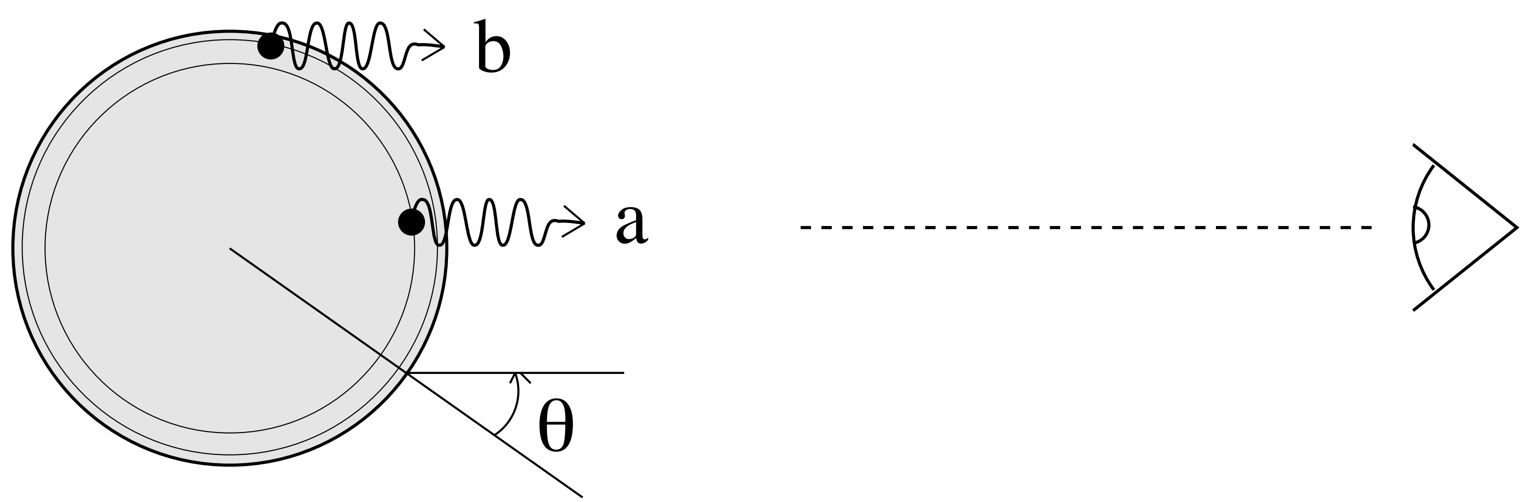
$$\boxed{I_\nu^+(0,\mu) \approx S_\nu(\tau_\nu = \mu)}$$

- Likewise for the emergent flux:

$$\begin{aligned} \mathcal{F}_\nu^+(0) &= 2\pi \int_0^1 \mu I_\nu^+(0,\mu) d\mu = \pi \sum_{n=0}^{\infty} \frac{2n!}{n+2} a_n \\ &= \pi \left(a_0 + \frac{2}{3} a_1 + a_2 + \cdots \right) \end{aligned}$$

$$\longrightarrow \quad \boxed{\mathcal{F}_\nu^+(0) \approx \pi S_\nu(\tau_\nu = 2/3)}$$

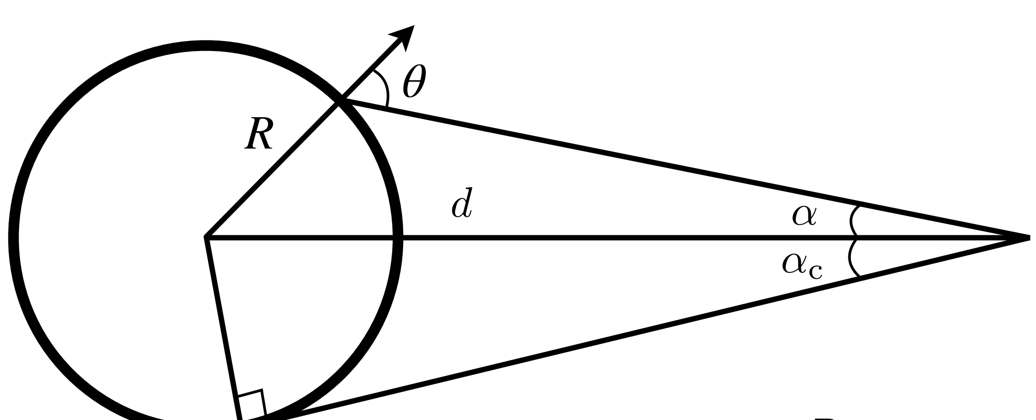
- Illustration



- Flux as measured by an observer:

① From the conservation of energy: $\boxed{F_\nu = \frac{4\pi R^2}{4\pi d^2}\mathcal{F}_\nu^+ \approx \frac{R^2}{d^2}\pi S_\nu(\tau_\nu = 2/3)}$ Here, R = stellar radius, and d = distance to the star.

② From the definition of flux: $F_\nu = 2\pi \int_0^{\alpha_c} I_\nu(0,\mu)\cos\alpha \sin\alpha d\alpha$



$\sin\alpha_c = \frac{R}{d}$

sine law: $\frac{\sin\alpha}{R} = \frac{\sin(\pi-\theta)}{d} = \frac{\sin\theta}{d}$

$$\Rightarrow \cos^2\alpha = \left(1 - \frac{R^2}{d^2}\right) + \frac{R^2}{d^2}\cos^2\theta$$

$$\cos\alpha \sin\alpha \, d\alpha = \frac{R^2}{d^2}\cos\theta \sin\theta \, d\theta$$

$$\therefore F_\nu = 2\pi\frac{R^2}{d^2}\int_0^{\pi/2} I_\nu(0,\mu)\cos\theta\sin\theta d\theta = \frac{R^2}{d^2}\mathcal{F}_\nu^+$$

Boltzmann distribution & Excitation Temperature

- **Boltzmann distribution**
  - a probability distribution that gives the probability that a system will be in a certain state as a function of that state's energy and the temperature of the system, when the system is in thermodynamic equilibrium.

$$p_i = \frac{N_i}{N} = \frac{g_i e^{-E_i/k_\mathrm{B}T}}{\sum_i g_i e^{-E_i/k_\mathrm{B}T}} \qquad \rightarrow \qquad \frac{N_j}{N_i} = \frac{g_j}{g_i} e^{-(E_j - E_i)/k_\mathrm{B}T}$$

$$\begin{aligned} g_i &= \text{degeneracy or statistical weight} \\ &= 2J+1 \text{ for a state with an angular momentum } J \end{aligned}$$

- **Excitation temperature**
  - Regardless of whether the system is in thermodynamic equilibrium (LTE) or not, the excitation temperature between two levels u and ℓ is defined as:

$$\frac{N_u}{N_\ell} = \frac{g_u}{g_\ell} e^{-(E_u - E_\ell)/k_\mathrm{B}T_\mathrm{exc}}$$

  - In general, the excitation temperatures between three levels, 1, 2, and 3, can all be different.

$$T_{12} \neq T_{13} \neq T_{23}$$

Transitions

Bound-bound Transitions

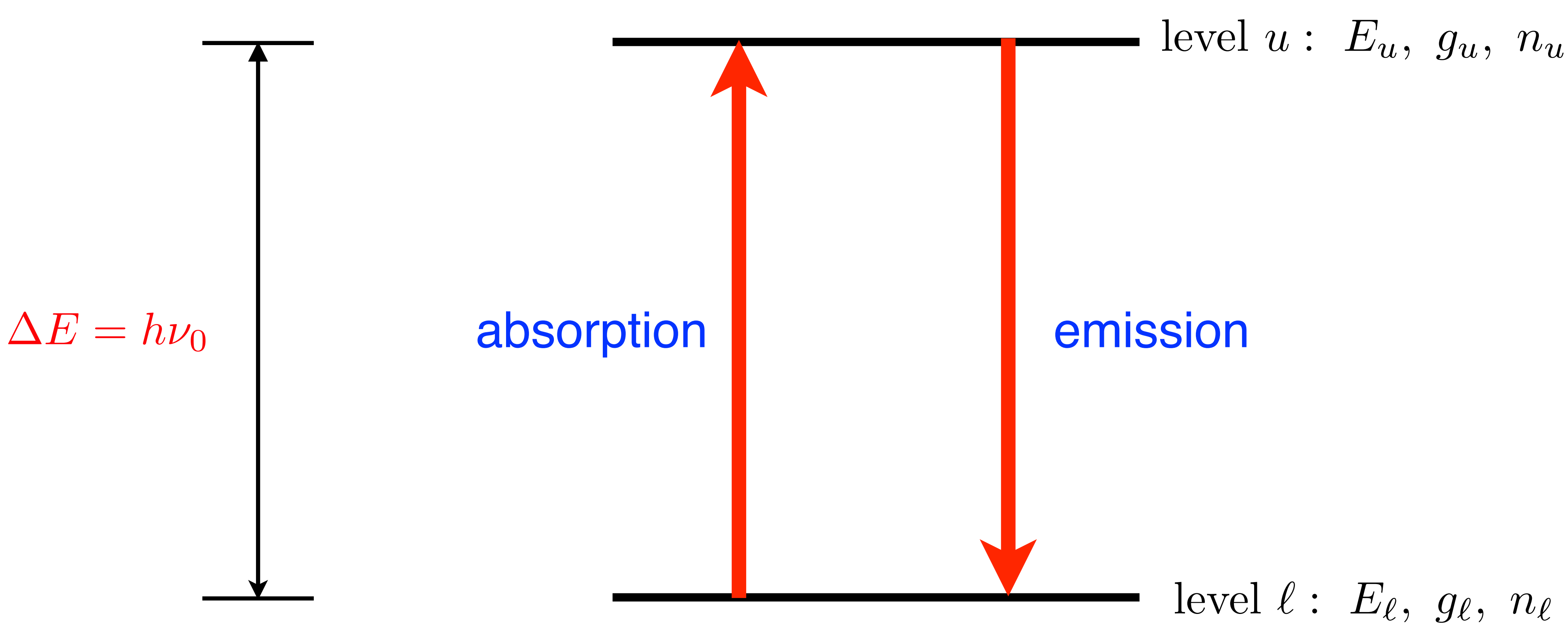
- **Excitation and de-excitation (Transition)**
  - Radiative excitation (photoexcitation; photoabsorption)
  - Radiative de-excitation (spontaneous emission; stimulated emission = induced emission)
  - Collisional excitation
  - Collisional de-excitation
- **Emission Line**
  - Collisionally-excited emission lines
  - Recombination lines (recombination following photoionization or collisional ionization)

Bound-free Transitions

- **Ionization**
  - Photoionization and Auger-ionization
  - Collisional Ionization (Direct ionization and Excitation-autoionization)
- **Recombination**
  - Radiative recombination <=> Photoionization
  - Dielectronic Recombination (not dielectric!)
  - Three-body recombination <=> Direct collisional ionization

Two Level System: The Einstein Coefficients

- Consider a system with two discrete energy levels (E_ℓ, E_u) and degeneracies (g_ℓ, g_u). Let (n_ℓ, n_u) be the number densities of atoms in levels (ℓ, u).

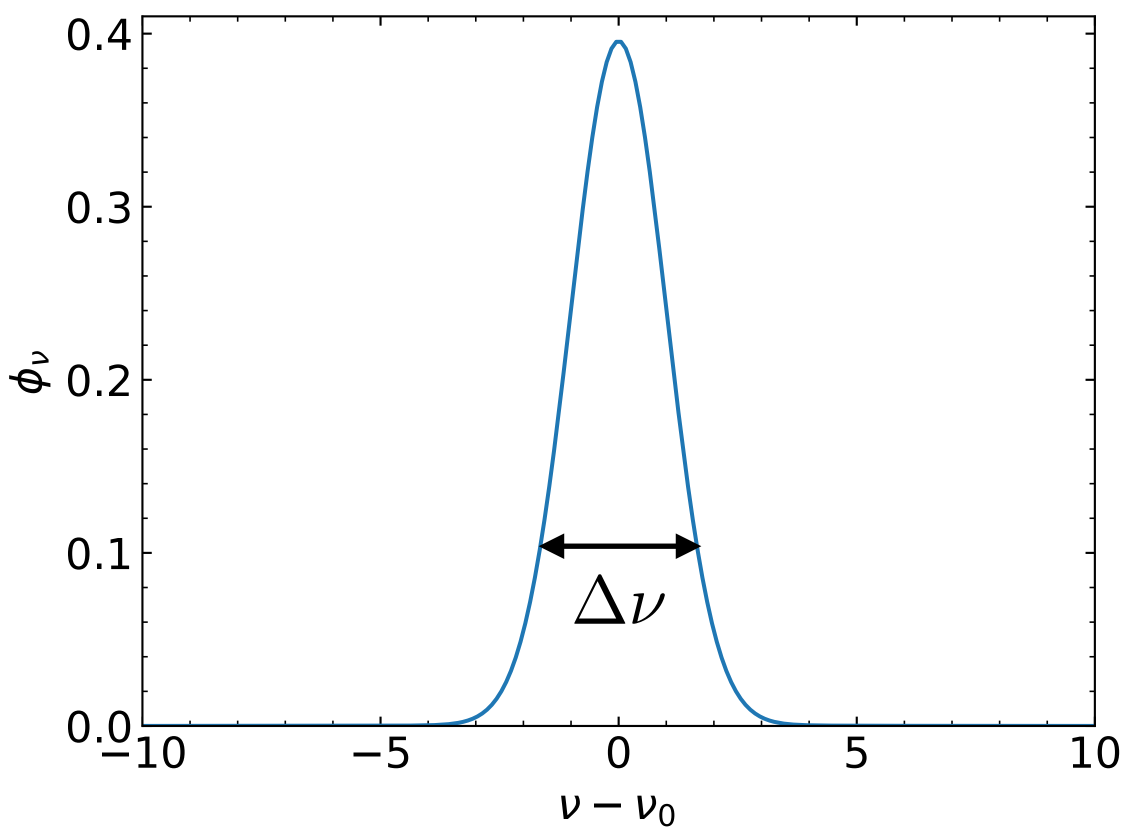


Here, g is the statistical weight, i.e., degeneracy of the level ($g = 2J + 1$).

- Two-level atoms are a useful idealization that permits detailed discussion of spatial non-locality due to photon scattering processes without having to bother with spectral non-locality due to photon conversion processes.

- For two-level atoms, the five (six) bound-bound processes maybe combined in different pairs:
  - **photon scattering:** radiative excitation followed by spontaneous or induced radiative deexcitation. The new photon is often described as still being the old one, but it has been re-directed and has possibly been slightly shifted (redistributed) in frequency.
  - **photon creation:** collisional excitation followed by spontaneous or induced radiative deexcitation. This pair makes a new photon out of kinetic energy.
  - **photon destruction:** radiative excitation followed by collisional deexcitation. This pair thermalizes a photon into kinetic energy.

- In general, the energy difference between the two levels have finite width which can be described by a line profile function $\phi(\nu)$.



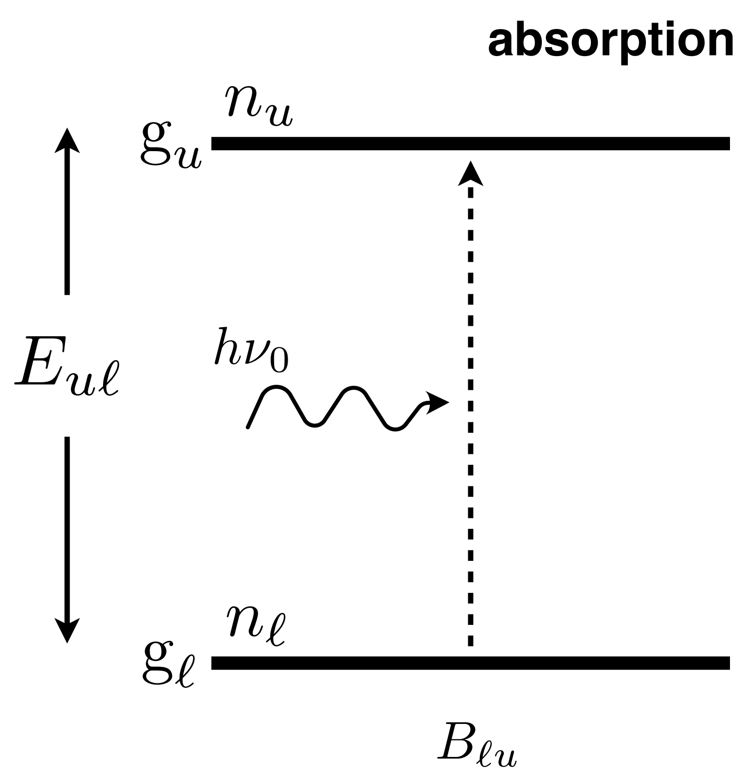
$$\int_0^\infty \phi_\nu d\nu = 1$$

The intensity is described by the mean over the line profile : $\bar{J} = \int J_\nu \phi_\nu d\nu$

Radiative Excitation and De-excitation (Absorption and Emission)

- **Three Radiative Transitions and Einstein Coefficients**
  - ***Absorption***:
    - If an absorber (atom, ion, molecule, or dust grain) X is in a lower level ℓ and there is radiation present with photons having an energy equal to $E_{u\ell}$. The absorber can absorb a photon and undergo an upward transition.

$$\text{absorption}: \quad X_\ell + h\nu_0 \;\rightarrow\; X_u \;(h\nu_0 = E_{u\ell})$$



    - The rate per volume at which the absorbers absorb photons will be proportional to both the (mean) intensity $\bar{J}$ of photons of the appropriate energy and the number density n_ℓ of absorbers in the lower level ℓ.

$$\left(\frac{dn_u}{dt}\right)_{\ell\rightarrow u} = -\left(\frac{dn_\ell}{dt}\right)_{\ell\rightarrow u} = n_\ell B_{\ell u}\bar{J}$$

    - The proportionality constant $B_{\ell u}$ is the ***Einstein B coefficient*** for the upward transition $\ell \rightarrow u$.

- ***Emission***:
  - An absorber X in an excited level u can decay to a lower level ℓ with emission of a photon. There are two ways this can happen:

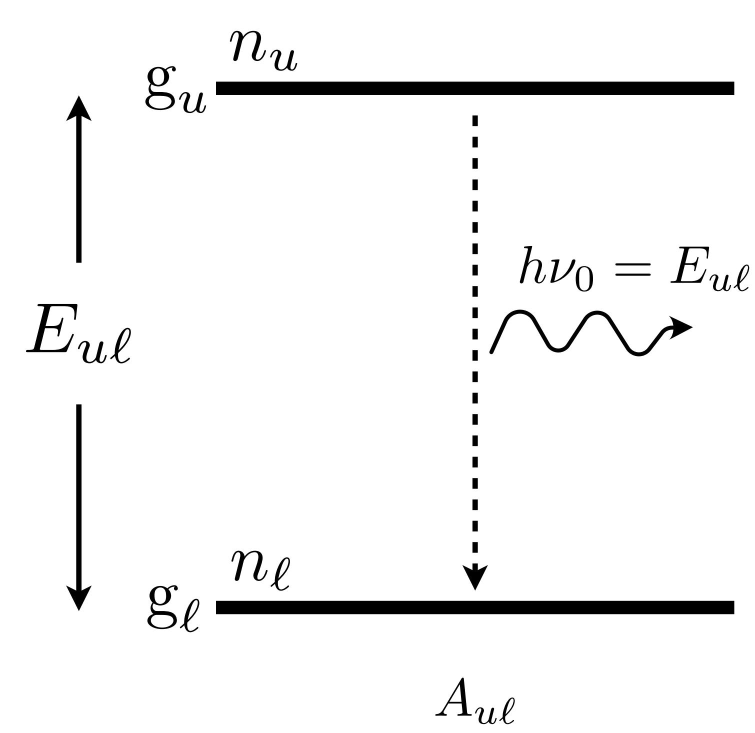
$$\text{spontaneous emission}: \quad X_u \;\rightarrow\; X_\ell + h\nu_0 \quad (h\nu_0 = E_{u\ell})$$
$$\text{stimulated emission}: \quad X_u + h\nu_0 \;\rightarrow\; X_\ell + 2h\nu_0 \quad (h\nu_0 = E_{u\ell})$$

  - ***Spontaneous emission*** is a random process, independent of the presence of a radiation field.
  - ***Stimulated emission*** occurs if photons of the identical frequency, polarization, and direction of propagation are already present, and the rate of stimulated emission is proportional to the intensity $\bar{J}$ of these photons.

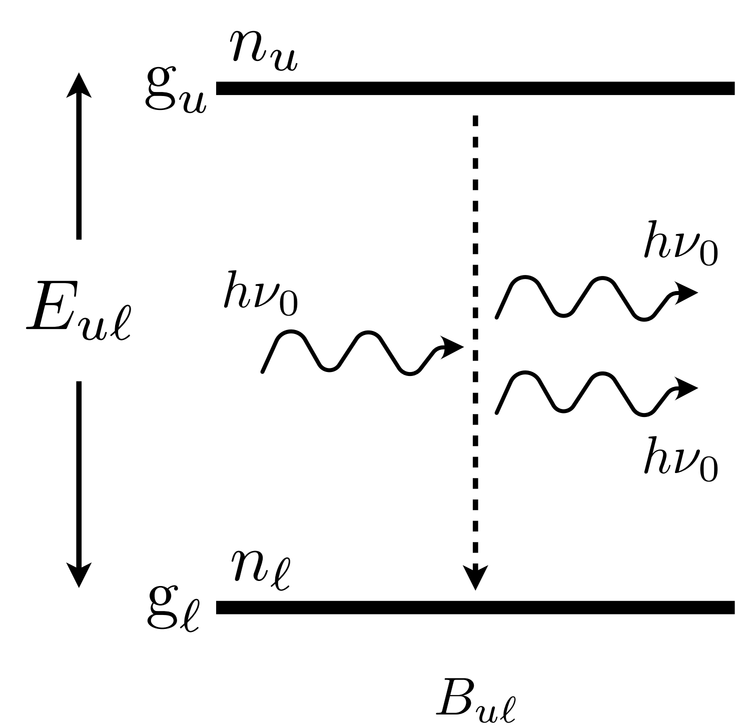
$$\left(\frac{dn_\ell}{dt}\right)_{u\rightarrow\ell} = -\left(\frac{dn_u}{dt}\right)_{u\rightarrow\ell} = n_u\left(A_{u\ell} + B_{u\ell}\bar{J}\right)$$

  - The probability per unit time $A_{u\ell}$ is the ***Einstein A coefficient*** for spontaneous transition. The coefficient $B_{u\ell}$ is the ***Einstein B coefficient*** for the downward transition $u \rightarrow \ell$.

spontaneous emission



stimulated emission



What is stimulated emission?

Stimulated emission from level u to level ℓ:

- The Einstein B-coefficient for stimulated emission is defined by

$B_{u\ell}\bar{J}$ = transition probability per unit time for stimulated emission.

- **Einstein found that to derive Planck's law another process was required that was proportional to radiation field and caused emission of a photon.**

- **The stimulated emission is precisely coherent (same direction and frequency, etc) with the photon that induced the emission.**

We also note that:

- **The energy density** is often used instead of intensity to define the Einstein B-coefficients, which leads to definitions differing by $c/4\pi$.

Absorption and Emission Coefficients in terms of Einstein coefficients

- The Einstein coefficients are useful means of analyzing absorption and emission processes. However, we often find it even more useful to use cross section because the cross section has a natural geometric meaning.
- ***(pure) Absorption cross section:***
  - The number density of photons per unit frequency interval is $u_\nu/h\nu = (4\pi/c)J_\nu/h\nu$. Let σ_ν be the cross section for absorption of photons for the transition $\ell \to u$. Then, the absorption rate is

$$\left(\frac{dn_u}{dt}\right)_{\ell\to u} = n_\ell \int d\nu \sigma_\nu \frac{4\pi J_\nu}{h\nu}$$

⇒ This should be the same as

$$n_\ell B_{\ell u}\bar{J} \equiv n_\ell B_{\ell u} \int J_\nu \varphi_\nu d\nu$$

  - Here, the cross section has a normalized profile of ϕ_ν. Then, we can write the absorption cross section as follows:

$$\sigma_\nu = \frac{h\nu}{4\pi} B_{\ell u}\varphi_\nu \quad \text{with} \quad \int \varphi_\nu d\nu = 1 \quad \text{and} \quad \int \sigma_\nu d\nu = \frac{h\nu_0}{4\pi} B_{\ell u}$$

because $\varphi_\nu = \varphi(\nu - \nu_0)$ is symmetric around $\nu = \nu_0$ in the absence of systematic Doppler shifts.

- ***Emission coefficient (Emissivity)***
  - The emissivity is defined as the power radiated per unit frequency per unit solid angle per unit volume.
  - The line emissivity can be expressed in terms of the spontaneous downward transition rate:

$$4\pi \int d\nu j_\nu = h\nu_0 \left(\frac{dn_\ell}{dt} \right)_{u\to\ell}^{\text{spontaneous}}$$

  - Comparing with the definition of the Einstein A coefficient, we obtain:

$$\int d\nu j_\nu = n_u \frac{A_{u\ell}}{4\pi} h\nu_0$$

  - If the emission line has a normalized profile of $\psi_\nu = \psi(\nu - \nu_0)$, we can write the emissivity as follows:

$$j_\nu = n_u \frac{A_{u\ell}}{4\pi} h\nu\psi_\nu \quad \text{with} \quad \int d\nu \psi_\nu = 1 \quad \text{and} \quad \int j_\nu d\nu = \frac{h\nu_0}{4\pi} n_u A_{u\ell}$$

because $\psi(\nu - \nu_0)$ is symmetric around $\nu = \nu_0$ in the absence of systematic Doppler shifts.

- ***(effective) Absorption Coefficient***
  - We note that the stimulated emission is proportional to the intensity of ambient radiation field. In the radiative transfer equation, it is convenient to include the stimulated emission term in the absorption coefficient as a negative absorption.

$$\left(\frac{dn_u}{dt}\right)_{\ell\to u} - \left(\frac{dn_\ell}{dt}\right)_{u\to\ell}^{\rm stimulated} = n_\ell B_{\ell u}\bar{J} - n_u B_{u\ell}\bar{J}$$

  - If we denote the profile shape for stimulated emission as $\chi_\nu = \chi(\nu-\nu_0)$, then the cross sections for spontaneous and stimulated emission can be written as follows:

$$\sigma_{\ell u} = \frac{h\nu}{4\pi}B_{\ell u}\varphi_\nu, \quad \text{and} \quad \sigma_{u\ell} = \frac{h\nu}{4\pi}B_{u\ell}\chi_\nu$$

  - **The net (effective) absorption coefficient is given by**

$$\begin{aligned}
\alpha_\nu^{\rm eff} &= n_\ell\sigma_{\ell u} - n_u\sigma_{u\ell} \\
&= \frac{h\nu}{4\pi}\left(n_\ell B_{\ell u}\varphi_\nu - n_u B_{u\ell}\chi_\nu\right) \\
&= \frac{h\nu}{4\pi}n_\ell B_{\ell u}\varphi_\nu\left(1 - \frac{n_u}{n_\ell}\frac{B_{u\ell}}{B_{\ell u}}\frac{\chi_\nu}{\varphi_\nu}\right)
\end{aligned}$$

- Source function and Relation
    - Using the Einstein relations, the absorption coefficient and source function can be written as follows:

$$S_\nu = \frac{j_\nu}{\alpha_\nu} = \frac{n_u A_{u\ell}\psi_\nu}{n_\ell B_{\ell u}\varphi_\nu - n_u B_{u\ell}\chi_\nu} = \frac{\frac{A_{u\ell}}{B_{u\ell}}\frac{\psi}{\varphi}}{\frac{n_\ell}{n_u}\frac{B_{\ell u}}{B_{u\ell}} - \frac{\chi}{\varphi}}$$

    - In LTE, the following two conditions should be satisfied.

$$(S_{\nu_0})_{\rm TE} = B_{\nu_0}(T) = \frac{2h\nu_0^3}{c^2}\frac{1}{e^{h\nu_0/k_{\rm B}T} - 1}$$

$$\left(\frac{n_u}{n_\ell}\right)_{\rm TE} = \frac{g_u}{g_\ell}e^{-E_{u\ell}/k_{\rm B}T} \quad \text{where } E_{u\ell} = h\nu_0.$$

    - Then, we found that the Einstein coefficients are related by the **Einstein relations** and that the three profiles are identical:

$A_{u\ell} = \frac{2h\nu_0^3}{c^2}B_{u\ell}$	$B_{\ell u} = \frac{g_u}{g_\ell}B_{u\ell}$	$\psi(\nu - \nu_0) = \varphi(\nu - \nu_0) = \chi(\nu - \nu_0)$

- The relations can be also derived for TE by equating the upward and downward radiative rates requiring detailed balance and equating the resulting expression for J_ν and B_ν at arbitrary temperature.
- They then hold universally since they do not depend on any medium property.

In summary, we obtained the following relations between the Einstein coefficients.

$$A_{u\ell} = \frac{2h\nu^3}{c^2} B_{u\ell} \qquad B_{u\ell} = \frac{c^2}{2h\nu^3} A_{u\ell}$$

$$B_{\ell u} = \frac{g_u}{g_\ell} B_{u\ell} \qquad B_{\ell u} = \frac{g_u}{g_\ell} \frac{c^2}{2h\nu^3} A_{u\ell}$$

- If we can determine any one of the coefficients, these relations allow us to determine the other two.
- These connect atomic properties (A_{ul}, B_{ul}, B_{lu}) and have no reference to the temperature. Thus, the relations must hold whether or not the atoms are in TE.
  - ✦ If the relations were only for TE, the relations would contain the dependence on T.
- Without stimulated emission, Einstein could not get Planck's law, but only Wien's law.
  - ✦ When $h\nu \gg k_B T$ (Wien's limit), level 2 is very sparsely populated relative to level 1. Then, stimulated emission is unimportant compared to absorption.

Generalized Kirchhoff's Law

- In general, the generalized Kirchhoff's law and the effective absorption coefficients are obtained as follows:

$$\boxed{\begin{aligned} & S_\nu = \frac{2h\nu^3}{c^2}\frac{1}{\frac{n_\ell/g_\ell}{n_u/g_u}-1} \quad \rightarrow \quad \text{generalized Kirchhoff's law} \\ & \alpha_\nu^{\text{eff}} = n_\ell\sigma_{\ell u}\left(1-\frac{n_u/g_u}{n_\ell/g_\ell}\right) = \frac{h\nu}{4\pi}n_\ell B_{\ell u}\varphi_\nu\left(1-\frac{n_u/g_u}{n_\ell/g_\ell}\right) \end{aligned}}$$

[Note] $$\frac{n_u/g_u}{n_\ell/g_\ell} = \exp\left(-E_{u\ell}/kT_{\text{exc}}\right)$$

- [Note] Complete redistribution
  - The complete redistribution holds when every process takes a fresh sample from the probability distribution, without "memory" of any preceding process. In this case, we have

$$\psi(\nu-\nu_0) = \varphi(\nu-\nu_0) = \chi(\nu-\nu_0)$$

Generalized Kirchhoff's Law

- ***Source Function***:

$$
\begin{aligned}
S_\nu &= \frac{j_\nu}{\alpha_\nu} \\
&= \frac{(1/4\pi) n_u A_{u\ell} h\nu_{u\ell} \phi_\nu^{\rm emiss}}{n_\ell \sigma_{\ell u}^{\rm eff}(\nu)} \\
&= \frac{n_u \frac{A_{u\ell}}{4\pi} h\nu_{u\ell} \phi_\nu^{\rm emiss}}{n_\ell \frac{g_u}{g_\ell} \frac{c^2}{8\pi\nu_{u\ell}^2} A_{u\ell} \phi_\nu^{\rm abs} \left[1 - \exp\left(-h\nu_{u\ell}/k_{\rm B}T_{\rm exc}\right)\right]} & \leftarrow \quad \frac{n_u}{n_\ell} = \frac{{\rm g}_u}{{\rm g}_\ell} \exp\left(-h\nu_{u\ell}/k_{\rm B}T_{\rm exc}\right) \\
&= \frac{2h\nu_{u\ell}^3}{c^2} \frac{1}{\exp\left(h\nu_{u\ell}/k_{\rm B}T_{\rm exc}\right) - 1} & \leftarrow \quad \phi_\nu^{\rm emiss} = \phi_\nu^{\rm abs}
\end{aligned}
$$

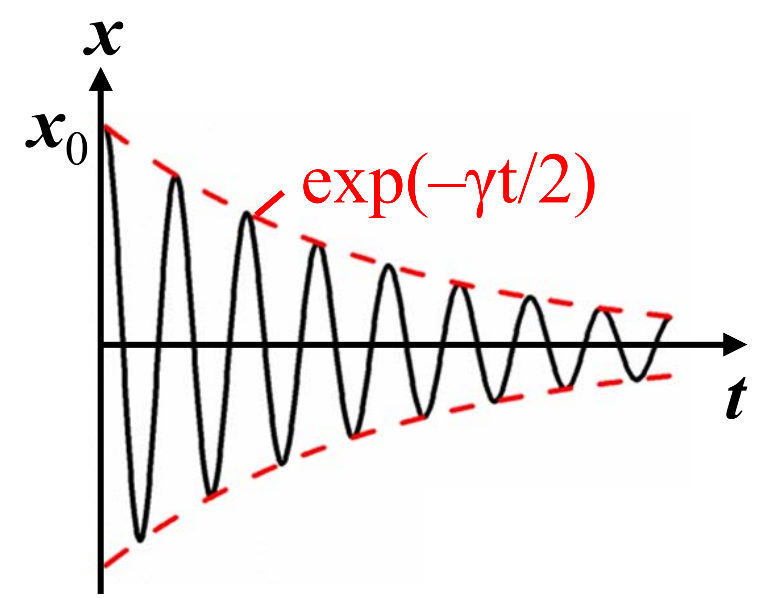
- This is called the ***generalized Kirchhoff's law***.
- **The intrinsic profiles for absorption and emission should are the same**.
  - The source function should approach the Planck function in LTE ($T_{\rm exc} = T_{\rm kinetic}$). For this to be true, the intrinsic profile of emission line should be the same as that of absorption line.
  - We can show that the intrinsic emission and absorption profiles are, indeed, the same, using a semi-classical model for an atom.

Intrinsic Line Profile - Lorentz Profile

[Classical Approach]

Suppose that the electric field energy decays in proportion to $e^{-\gamma t}$ and then the electric field is of the form $e^{-\gamma t/2}$.

Here, γ is the damping constant (damping per unit time).



We then have an emitted spectrum determined by the decaying sinusoid type of electric field.

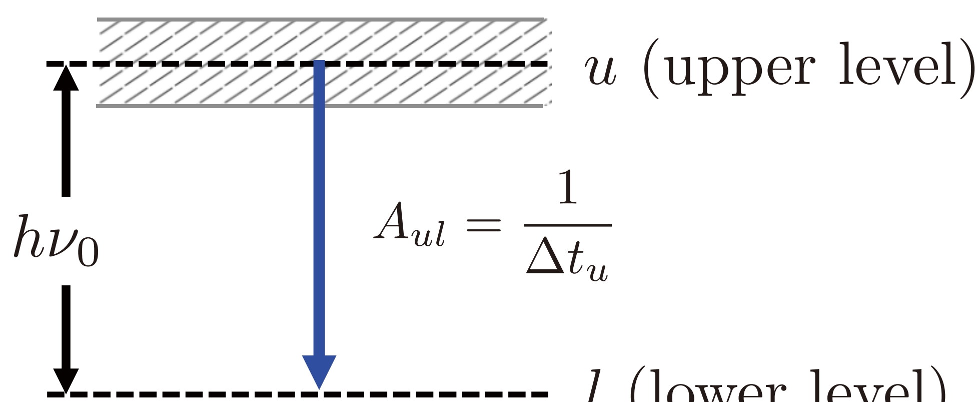
Its Fourier transform (spectral profile) is a Lorentz (or natural, or Cauchy) profile:

$$\boxed{\phi_\nu = \frac{\gamma/4\pi^2}{(\nu - \nu_0)^2 + (\gamma/4\pi)^2} \qquad \text{[unit: per Hz]}}$$

Intrinsic Line Profile - Einstein A coefficient

- **Natural Broadening**
  - The intrinsic line width of a line is due to ***the Heisenberg uncertainty principle.*** If an energy level u has a lifetime Δt, then uncertainty (spread) in energy ΔE must be $\Delta E \sim \hbar/\Delta t$ ($\hbar = h/2\pi$), and the resulting spread in the frequency of emitted photons is $\Delta\nu = \Delta E/h$.

(1) Line width due to the uncertainty principle:



A_{ul} = decay rate
= decay probability per unit time, Einstein A coefficient.

$$\Delta E_u = \text{uncertainty in energy of } u$$
$$\Delta t_u = \text{the uncertainty in time of occupation of } u$$
$$\Delta\nu_u = \text{spread in frequency}$$
$$= \Delta E_u/h = 1/(2\pi\Delta t_u) = A_{ul}/(2\pi)$$

(2) Line width of the Lorentz function:

$$\phi_\nu = \frac{\gamma/4\pi^2}{(\nu-\nu_0)^2 + (\gamma/4\pi)^2}$$ [unit: per Hz]

In terms of the line width $\Delta\nu$, the normalized Lorentz profile can be rewritten as

$$\boxed{\phi_\nu = \frac{1}{2\pi}\frac{\Delta\nu/2}{(\nu-\nu_0)^2 + (\Delta\nu/2)^2}}$$

Hence, the FWHM of the Lorentz function: $\Delta\nu_u = \gamma/2\pi$

Comparing (1) and (2), we find that γ is equivalent to the the Einstein A-coefficient., i.e., $\gamma = A_{ul}$.

Semiclassical (Weisskopf-Woolley) Picture of Quantum Levels

- In the semiclassical picture, each level is viewed as a continuous distribution of sublevels with energies close to the energy of the level (E_n).

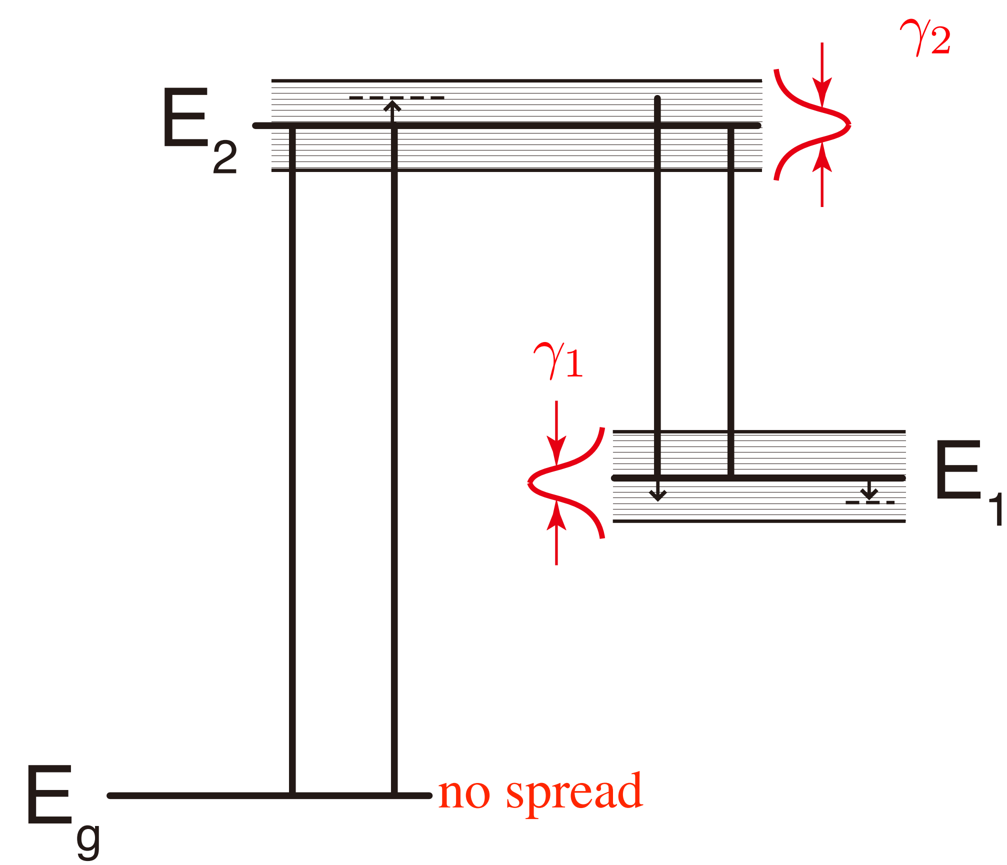
  The distribution of sublevels are explained by the Heisenberg Uncertainty Principle. The level has a lifetime $\Delta t = 1/A$ (A = Einstein A coefficient) and a spread in energy about $\Delta E \approx \hbar/\Delta t = \hbar A$.

$$\Delta E \Delta t \approx \hbar$$

The ground level has no spread in energy

because $\Delta t = \infty$.

The atom is in a definite sublevel of some level.



A transition in a spectral line is considered to be an **instantaneous transition** between a definite sublevel of an initial level to a definite sublevel of a final level.

- ***The energy spread of sublevels is described by a Lorentzian profile with the damping parameter of $\gamma = A$.***
- ***This picture implies that the emission line profile is the same as the absorption line profile.***

Photon Occupation Number

- We note the Einstein coefficients are intrinsic properties of the absorbing material, irrelevant to the assumption of TE. Hence, ***the relations between the Einstein coefficients should hold in any condition.***
- Using the relation, we can rewrite the downward and upward transition rates:

$$\left(\frac{dn_u}{dt}\right)_{\ell\to u} = n_\ell \frac{g_u}{g_\ell}\frac{c^2}{2h\nu^3}A_{u\ell}J_\nu \qquad \left(\frac{dn_\ell}{dt}\right)_{u\to\ell} = n_u A_{u\ell}\left(1+\frac{c^2}{2h\nu^3}J_\nu\right)$$

- Sometimes, it is helpful to use a dimensionless quantity, the ***photon occupation number***:

$$n_\gamma \equiv \frac{c^2}{2h\nu^3}I_\nu \quad \xrightarrow{\text{averaging over directions}} \quad \langle n_\gamma\rangle = \frac{c^2}{2h\nu^3}J_\nu = \frac{1}{e^{h\nu/k_{\rm B}T}-1}$$

- Then, the above transition rates are simplified:

$$\left(\frac{dn_u}{dt}\right)_{\ell\to u} = n_\ell \frac{g_u}{g_\ell}A_{u\ell}\langle n_\gamma\rangle \qquad \left(\frac{dn_\ell}{dt}\right)_{u\to\ell} = n_u A_{u\ell}\left(1+\langle n_\gamma\rangle\right)$$

- The photon occupation number determines the relative importance of stimulated and spontaneous emission: stimulated emission is important only when $\langle n_\gamma\rangle \gg 1$.

- The correction factor for the stimulated emission in absorption coefficient:
  - For Lyα line,

$$h\nu_{u\ell} = 10.2\,\mathrm{eV} \quad \rightarrow \quad 1 - \exp\left(-\frac{h\nu_{u\ell}}{k_{\mathrm{B}}T_{\mathrm{exc}}}\right) = 1 - \exp\left(-\frac{1.1837\times10^5\,\mathrm{K}}{T_{\mathrm{exc}}}\right)$$
$$\simeq 1 \quad \text{for } T_{\mathrm{exc}} \approx T_{\mathrm{gas}} < 1\times10^5\,\mathrm{K}$$

    - The stimulated emission is negligible.
  - For 21 cm line,

$$h\nu_{u\ell} = 6\,\mu\mathrm{eV} \quad \rightarrow \quad 1 - \exp\left(-\frac{h\nu_{u\ell}}{k_{\mathrm{B}}T_{\mathrm{exc}}}\right) = 1 - \exp\left(-\frac{0.068\,\mathrm{K}}{T_{\mathrm{exc}}}\right)$$
$$\simeq \frac{0.068\,\mathrm{K}}{T_{\mathrm{exc}}} \ll 1 \quad \text{for } T_{\mathrm{exc}} \approx T_{\mathrm{gas}} \sim 100\,\mathrm{K}$$

    - The correction for stimulated emission is very important. ***We, therefore, need to take into account the stimulated emission in dealing with the 21 cm line.***

- Two limiting cases:
  - At radio and sub-mm frequencies, the upper levels are often appreciably populated, and it is important to include both spontaneous and stimulated emission.
  - When we consider propagation of optical, UV, or X-ray radiation in cold ISM, the upper levels of atoms and ions usually have negligible populations, and stimulated emission can be neglected.

Oscillator Strength

- In the previous slides, we characterized the absorption cross section by the Einstein A coefficient. Equivalently, we can express the cross section in terms of the oscillator strength for the absorption transition $\ell \rightarrow u$, defined by the relation:

$e = 4.803 \times 10^{-10} \mathrm{esu}\ (= \mathrm{cm}^{3/2}\, \mathrm{g}^{1/2}\, \mathrm{s}^{-1})$

$$\int \sigma_{\ell u}(\nu) d\nu = \frac{\pi e^2}{m_e c} f_{\ell u} \quad \rightarrow \quad \boxed{\sigma_{\ell u}(\nu) = \frac{\pi e^2}{m_e c} f_{\ell u} \phi_\nu = 0.02654\, f_{\ell u} \phi_\nu \ \mathrm{cm}^2}$$

  - Here, the factor $\frac{\pi e^2}{m_e c} = 0.02654\ \mathrm{cm}^2\,\mathrm{Hz}$ is the cross-section, integrated over the line profile, for a classical oscillator model.
  - The oscillator strength is the factor which corrects the classical result. The quantum mechanical process can be interpreted as being due to a (fractional) number f of equivalent classical electron oscillators of the same frequency.

- The Einstein A coefficient is related to the absorption oscillator strength of the upward transition by

$$\boxed{A_{u\ell} = \frac{8\pi^2 e^2 \nu_{u\ell}^2}{m_e c^3} \frac{\mathrm{g}_\ell}{\mathrm{g}_u} f_{\ell u} = \left(\frac{0.8167\,\mathrm{cm}}{\lambda_{u\ell}} \right)^2 \frac{\mathrm{g}_\ell}{\mathrm{g}_u} f_{\ell u} \ [\mathrm{s}^{-1}]}$$

  - For 21.1 cm line, $\mathrm{g}_u = 3,\ g_\ell = 1\ (g_F = 2F + 1)$

$$A_{u\ell} = 2.88 \times 10^{-15}\ [\mathrm{s}^{-1}] = (11\,\mathrm{Myr})^{-1} \qquad f_{\ell u} = 5.75 \times 10^{-12}$$

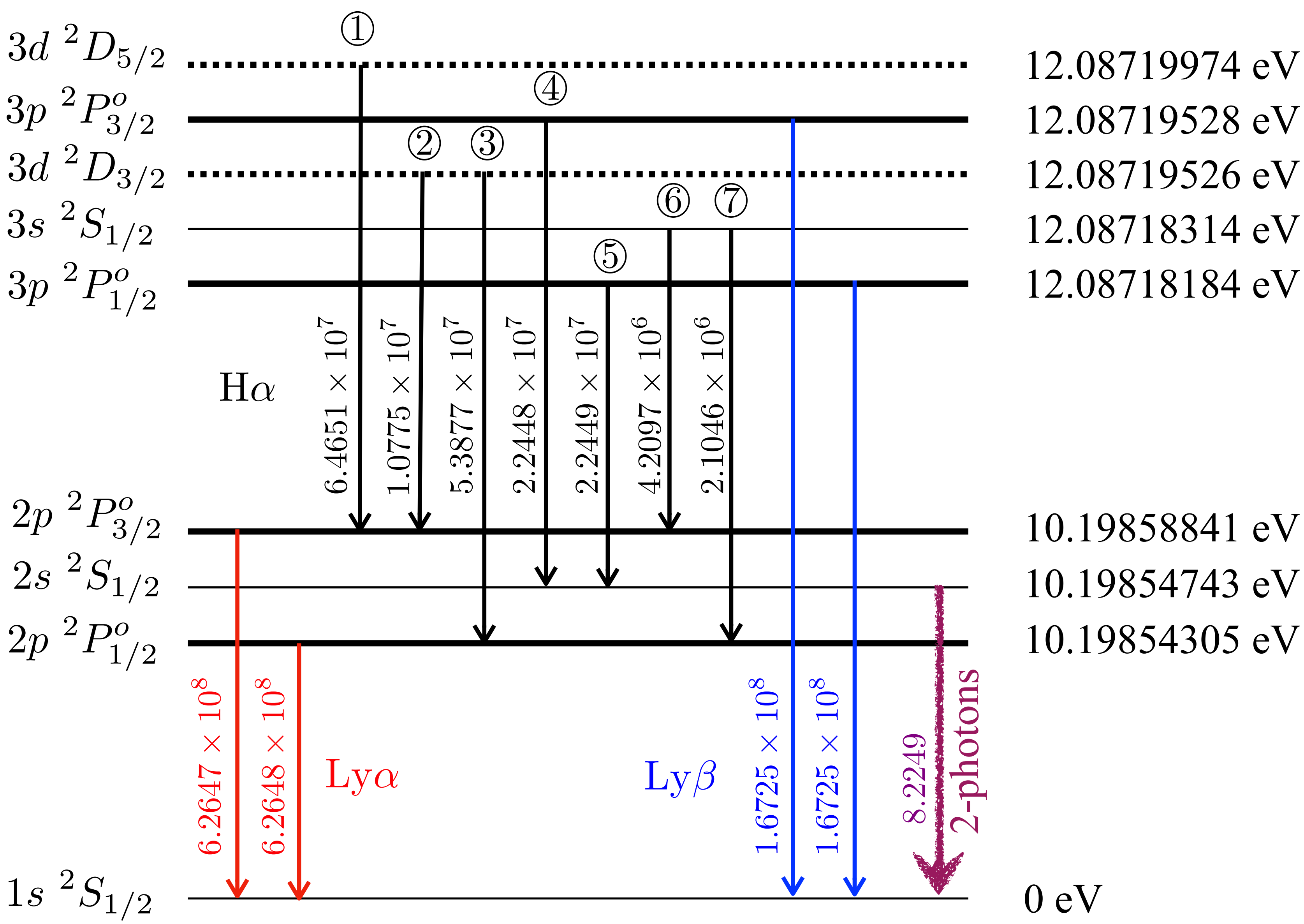
  - For Lyα (1215.67Å) line, $\mathrm{g}_u = 3,\ g_\ell = 1\ (g_L = 2L + 1)$

$$A_{u\ell} = 6.265 \times 10^8\ [\mathrm{s}^{-1}] \quad f_{\ell u} = 0.4164 \ \text{ for } 1^2S_{1/2} \rightarrow 2^2P$$

$$f_{\ell u} = 0.27760 \text{ for } {}^2S_{1/2} \rightarrow {}^2P_{3/2}$$
$$= 0.13881 \text{ for } {}^2S_{1/2} \rightarrow {}^2P_{1/2}$$

Einstein A coefficients for Hydrogen Atom

Energy Levels and Damping Constants (s^{-1})



Energy Levels of Neutral Hydrogen (NIST)

Configuration	Term	J	Level (cm^{-1})	Ref.
1*s*	^2S	1/2	0.0000	MK00a
2*p*	$^2P°$	1/2	82258.9191	MK00a
		3/2	82259.2850	MK00a
2*s*	^2S	1/2	82258.9544	MK00a
3*p*	$^2P°$	1/2	97492.2112	MK00a
		3/2	97492.3196	MK00a
3*s*	^2S	1/2	97492.2217	MK00a
3*d*	^2D	3/2	97492.3195	MK00a
		5/2	97492.3556	MK00a
4*p*	$^2P°$	1/2	102823.8486	MK00a
		3/2	102823.8943	MK00a
4*s*	^2S	1/2	102823.8530	MK00a
4*d*	^2D	3/2	102823.8942	MK00a
		5/2	102823.9095	MK00a
4*f*	$^2F°$	5/2	102823.9095	MK00a
		7/2	102823.9171	MK00a
5*p*	$^2P°$	1/2	105291.6287	MK00a
		3/2	105291.6521	MK00a
5*s*	^2S	1/2	105291.6309	MK00a
5*d*	^2D	3/2	105291.6520	MK00a
		5/2	105291.6599	MK00a
5*f*	$^2F°$	5/2	105291.6598	MK00a
		7/2	105291.6637	MK00a
5*g*	^2G	7/2	105291.6637	MK00a
		9/2	105291.6661	MK00a
H	***Limit***		**109678.7717**	MK00a

Continuum Transitions

- Inelastic processes
  - Bound-free transitions (photoionization)
  - Free-free transitions

- Elastic processes
  - Thomson scattering
  - Rayleigh scattering

- Inelastic processes
  - **[Bound-free transitions]** The cross section (per particle) for bound-free transitions of hydrogen and hydrogen-like ions is given by Kramers' formula (Kramers 1923; Gaunt 1930):

$$\sigma_\nu^{\mathrm{bf}} = \frac{32}{3^{3/2}} \frac{\pi^2 e^6}{h^3} \frac{R_\infty}{n^5 \nu^3} g_{\mathrm{bf}} = 2.815 \times 10^{29} \frac{Z^4}{n^5 \nu^3} g_{\mathrm{bf}} \ [\mathrm{cm}^2] \quad \text{for } \nu \geq \nu_0$$

$$g_{\mathrm{bf}} = 1 - \frac{0.3456}{(\lambda R_\infty)^{1/3}} \left(\frac{\lambda R_\infty}{n^2} - \frac{1}{2} \right)$$

⇒ adequate for most purposes (Menzel & Pekeris 1935)

Here, n is the principal quantum number of the level i from which the atom or ion is ionized, Z the ion charge, ν_0 the threshold frequency, ν in Hz and g_{bf} is the dimensionless Gaunt factor.

  - We have to take into account the populations in each level.

    In LTE, the population is given by

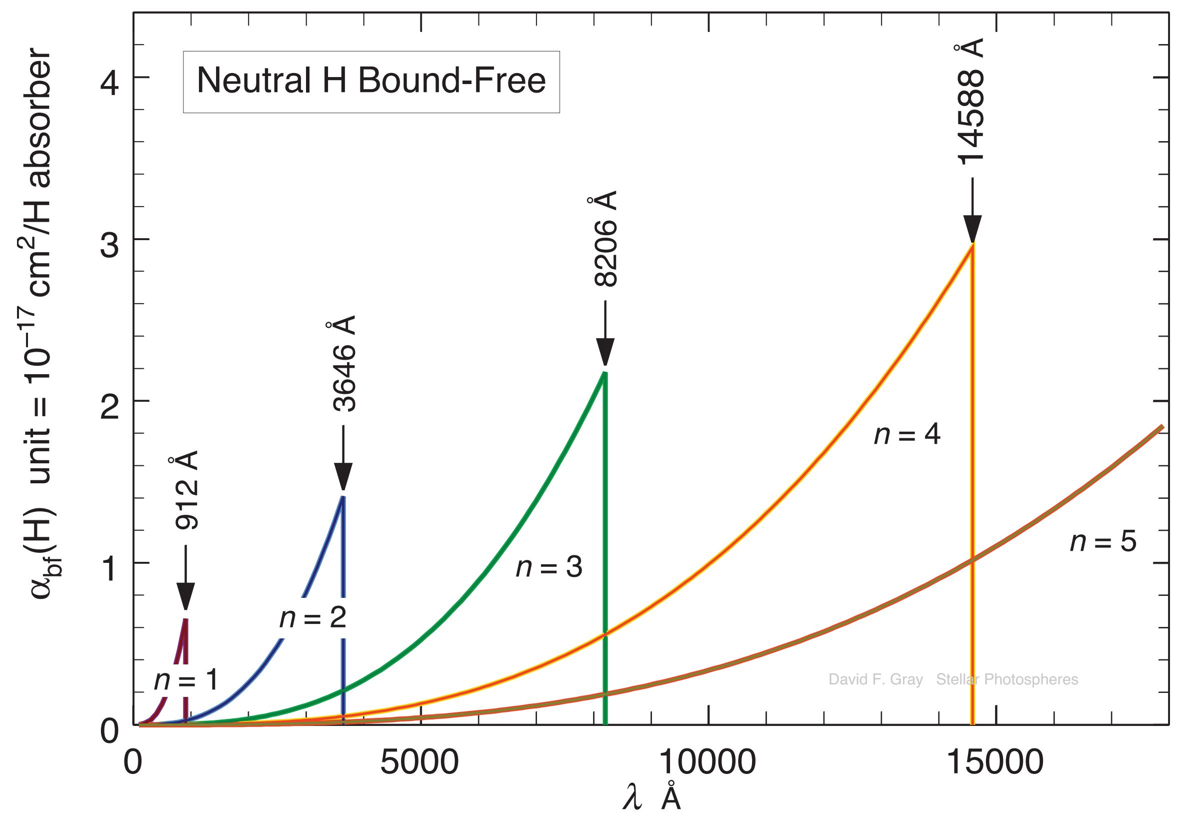
$$\frac{N_n}{N} = \frac{g_n}{u(T)} e^{-\chi_n/kT}$$

$$g_n = 2n^2 \qquad \text{statistical weight}$$

$$u(T) = 2 \qquad \text{partition function}$$

$$\chi_n = hcR\left(1 - 1/n^2\right) = 1.3598\left(1 - 1/n^2\right)\ \mathrm{eV}$$

excitation potential for the bound levels



- The cross section for neutral H atom

$$\sigma_{\rm bf}^{\rm tot}(\mathrm{H}) = \sum_{n=1}^{\infty} \sigma_{\rm bf}(\mathrm{H}) \frac{N_n}{N}$$

In the low density medium, such as the ISM, $n_n \approx 0$ for $n \geq 2$.

- For more complex atoms and ions, the bound-free cross sections do not have such simple ν^{-3} dependence but possess peaks at "resonances" caused by other electrons.

- **[Free-free transitions/absorption]**
- The free-free absorption of neutral hydrogen is much smaller than the bound-free. When the free electron has a collision with a proton, its trajectory is altered. If a photon is absorbed during such a collision, the energy of the electron is increased by the photon energy. The strength of the absorption depends on the electron velocity v, because a slow encounter increases the probability of a photon passing by during the collision.
- The cross section of free-free transitions for an ion with charge $Z(\geq 0)$ is

$$\sigma_\nu^{\rm ff} = \frac{2}{3^{3/2}} \frac{h^2 e^2 R_\infty}{\pi m_e^3} \frac{1}{\nu^3} \left(\frac{2m_e}{\pi k T} \right)^{1/2} g_{\rm ff} \quad \Leftarrow \quad R_\infty = \frac{2\pi^2 m_e e^4}{ch^3} = 1.097373 \times 10^5 \,\mathrm{cm}^{-1}$$

$$= 3.7 \times 10^8 \; n_e \frac{Z^2}{T^{1/2} \nu^3} g_{\rm ff}$$

$$g_{\rm ff} = 1 + \frac{0.3456}{(\lambda R_\infty)^{1/3}} \left(\frac{\lambda k T}{hc} + \frac{1}{2} \right)$$

⇒ given by Menzel & Pekeris (1935)
See also van Hoof et al. (2014, MNRAS, 444, 420)

- The absorption coefficient is then

$$\alpha_\nu^{\rm ff} = \sigma_\nu^{\rm ff} \left(1 - e^{-h\nu/kT}\right) n_{\rm ion}$$

which holds as long as the velocities are Maxwellian. The negative term corrects for induced processes.

- Approximations

$$\alpha_\nu^{\rm ff} \approx 3.7 \times 10^8 n_e n_{\rm ion} \frac{Z^2}{T^{1/2}\nu^3} g_{\rm ff} \qquad \text{in the Wien limit.}$$

$$\approx 0.018 n_e n_{\rm ion} \frac{Z^2}{T^{3/2}\nu^2} g_{\rm ff} \qquad \text{in the Rayleigh-Jeans limit.}$$

- Elastic processes
  - **[Thomson scattering]** Thomson scattering of photons by free electrons has a frequency-independent cross section:

$$\sigma^{\mathrm{T}} = \frac{8\pi}{3} r_e^2 = 6.648 \times 10^{-25} \mathrm{cm}^2, \quad r_e = \frac{e^2}{m_e c^2} \quad \text{(classical electron radius)}$$

  - **[Rayleigh scattering]** The cross section for Rayleigh scattering of photons with $\nu \ll \nu_0$ by bound electrons with bounding energy $h\nu_0$ is:

$$\sigma_\nu^{\mathrm{R}} \approx f_{\ell u} \sigma^{\mathrm{T}} \left(\frac{\nu}{\nu_0} \right)^4 = f_{\ell u} \sigma^{\mathrm{T}} \left(\frac{\lambda_0}{\lambda} \right)^4$$

  - Here, f_{lu} is the oscillator strength of the corresponding bound-bound “resonance transition” of the bound electron, for example the Lyα transition in neutral hydrogen or a weighted sum over all Lyman lines.
  - The λ^{-4} dependence makes our sky blue and sunsets red.
  - **[Scattering phase function]** Thomson and Rayleigh scattering are coherent, meaning elastic; the photon gets redirected but it keeps its frequency. Their redirection phase function has the form of $1 + \cos^2\theta$. This cross section hold for low-energy photons and low-energy electrons. For high-energy photons, Thomson scattering is replaced by Compton scattering; for high-energy electrons, by inverse Compton scattering.